

LLM2025	MentalMate	CC1
Coordinated by:	VU Anh Duy	2 months
Course : Theory and Practical Applications of LLMs		

MentalMate: A Multimodal LLM-Based Mental Wellness Chatbot Integrating Wearable Sensor Data

I Context, Positioning and Objectives

a Context and State of the Art

Mental health is a global crisis: 800,000 suicide deaths annually and 85% of people in low-middle income countries receive no treatment [1, 10]. This gap drives interest in AI chatbots as scalable solutions.

LLMs like GPT-4 and Llama-2 generate responses rated 10x more empathic than physicians [12, 14]. However, current chatbots face limitations: **(1)** no physiological awareness; **(2)** generic, non-personalized responses; **(3)** prompt engineering rather than fine-tuning [18, 20]; **(4)** safety risks from hallucinations [3, 11]. Research confirms stress manifests physiologically before verbal expression [13], and reviews show evolution from rule-based to LLM systems (45% in 2024) [5, 15, 16].

Existing Multimodal Systems: EMLLM [4] uses Falcon-7B for workplace stress management. HEALTH-LLM [7] fine-tunes on WESAD for health prediction. PHYSIOLLM [9] uses Fitbit for health insights. However, **these focus on prediction, not therapeutic dialogue.**

Emotional Support: ESCONV [8] provides 1,053 dialogues with 8 support strategies from Helping Skills Theory (Exploration → Comforting → Action) [2, 17, 19, 20]. Wysa [6] demonstrates commercial feasibility.

Project Goal: MENTALMATE fine-tunes on ESCONV strategies + WESAD sensors for empathetic, strategy-aware therapeutic dialogue. **Differentiators:** (i) 8 therapeutic strategies; (ii) sensor-informed strategy selection; (iii) empathy-first architecture; (iv) structured multi-turn support.

b Methodology and Objectives

b1 Obj 1: Fine-Tune Multimodal LLM with Sensor Integration

We fine-tune Llama-2-7B using **LoRA (Rank $r=8$, $\alpha=16$)** targeting attention layers (q_proj , v_proj) for parameter-efficient adaptation (1-5% parameters).

- **Data:** ESCONV (1,053 conversations with strategy annotations and empathy ratings) combined with WESAD physiological features (HR, HRV, EDA, temperature from 15 subjects across neutral/stress/amusement states).
- **Training:** Cross-entropy loss augmented with strategy-aware objective (weighted cross-entropy over 8 strategies) encouraging appropriate support strategy selection based on conversation stage and physiological context. Training: 3 epochs, batch size 4, learning rate $2e-4$.

b2 Obj 2: Sensor-Text Fusion Architecture

We design a fusion architecture enabling the LLM to condition responses on physiological states for context-aware generation.

- **Encoder:** A 3-layer feedforward network (ReLU activations) transforms raw sensor signals into 768-dim embeddings matching the LLM embedding space.
- **Fusion:** **Early fusion** strategy prepending sensor embeddings as special tokens (`[HIGH_STRESS]`, `[POOR_SLEEP]`, `[LOW_ACTIVITY]`) to conversation history. A temporal attention mechanism weights recent readings (24h) vs. historical patterns (7-day average) to distinguish acute from chronic stress.
- **Validation:** Ablation studies comparing early fusion, late fusion, and cross-attention approaches measuring correlation between sensor states and response characteristics (strategy selection, empathy level).

b3 Obj 3: Empathy-Aware Generation & Safety

We implement constrained decoding and validation ensuring therapeutic quality and critical safety boundaries.

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- **Empathy:** Empathy prediction head (linear layer) optimized alongside language modeling using ES-CONV’s empathy annotations (1-5 scale), encouraging high-empathy response generation.
- **Strategy Constraint:** Controlled decoding biasing token probabilities toward appropriate strategies per conversation stage (Exploration → Questions; Comforting → Affirmation; Action → Providing Suggestions).
- **Safety:** Dual-filter system: (a) rule-based keyword detection for crisis indicators, (b) RoBERTa classifier for mental health crisis detection. Intervention triggers override generation with crisis resources when severe distress detected.

b4 Obj 4: Comprehensive Evaluation

We evaluate on held-out test set (15% of ESConv) plus synthetic scenarios combining diverse sensor states with conversation contexts.

- **Automatic Metrics:** BLEU/ROUGE (text quality), Perplexity (language modeling), Distinct-1/2 (diversity).
- **Empathy Evaluation:** RoBERTa-based classifier scoring Cognitive Empathy (understanding) and Affective Empathy (emotional resonance) on generated responses.
- **Sensor Conditioning:** Quantitative analysis of correlation between sensor states and response characteristics (high stress → more Affirmation strategies; poor sleep → sleep hygiene suggestions).
- **Baselines:** (a) Vanilla Llama-2-7B (zero-shot), (b) Text-only fine-tuned (ESConv without sensors), (c) GPT-3.5 with prompt engineering, (d) EmLLM-style prediction-focused approach.
- **Ablation Studies:** Impact of sensor integration, LoRA rank ($r=4,8,16$), training data size, fusion strategy.

II Timeline (2-Month Execution)

Given the accelerated 2-month timeline (Nov 2024 – Jan 2025), the project is structured into four overlapping Work Packages (WPs) to ensure the multimodal fusion architecture is trained and validated before the final deadline.

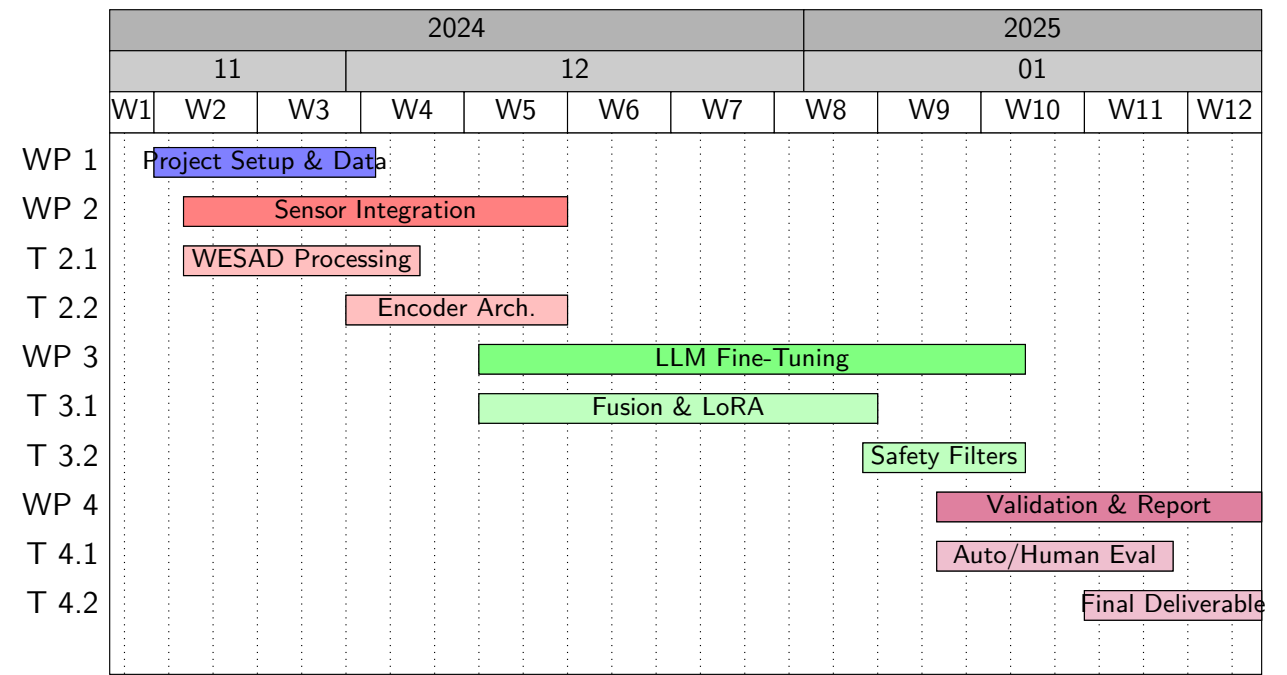


Figure 1. 2-Month Project Schedule (Nov 2024 - Jan 2025)

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